Auditory Superstitious-Perception: results walkthrough

Contents

1 The experiment

26 college students completed two blocks of 149 trials each. On every trial they heard a short noisy audio clip and reported whether it was the "target" or the "distractor". The underlying target and distractor clips are essentially noise; chance performance is 0.50.

The task revolves around listening for a target word ("wall") in the context of a target sentence ("The picture hung on the wall"). We cut out the final word ("wall") and the stimuli are white noise audio clips with the same length (370ms), sampling rate (8000Hz), and channel count (1- Mono) as the extracted "wall" clip.

There are target stimuli and distractor stimuli. Target stimuli are more correlated (according to Pearson R) with the "wall" clip than the distractor stimuli are. The question is whether, and under what circumstances, subjects can pick up on the correlation between the target stimuli and the missing word, and correctly categorize white noise stimuli into target vs. distractor.

There were two block types. These are labelled `full_sentence` and `imagined_sentence`. In the `full_sentence` block subjects hear the beginning of the target sentence ("The picture hung on the") followed by one of the white noise stimuli on each trial. In the `imagined_sentence` blocks subjects must imagine the beginning of the sentence (e.g. "The picture hung on the wall") and then click a button to hear the white noise stimulus.

Subjects were recruited at Oberlin ($n = 16$) and UChicago ($n = 10$). Every comparison is reported three ways: each site alone and pooled ("Unified").

2 Brief jargon primer

Skip if you already know what a p-value, Cohen's d, and cosine similarity are.

2.1 Accuracy and chance

Considering each subject's performance as a proportion of stimuli called correctly between 0 and 1 calculated via $\frac{\textbf{Correct Responses}}{\textbf{Total Stimuli Presented}}$. If subjects were guessing or at chance, then we would expect scores to cluster around 0.50.

The intuition: imagine flipping a fair coin 150 times and calling each flip. You will get close to 75 right but almost never exactly 75. By chance alone you might land at 80/150 (0.533) or 70/150 (0.467). That random spread around 0.50 is what a single subject looks like under the null. The question for each group is whether the cloud of subject means sits noticeably to the right of 0.50 *more often than that random spread would explain.*

Each subject completed $N = 149$ trials, with either 75 targets and 74 distractors or vice versa. Under a guessing model with no systematic knowledge (i.e., $d' = 0$), but allowing an arbitrary response bias $b = P(\text{“target”})$, the expected accuracy is

$$E[\text{accuracy}] = \frac{Kb + (N - K)(1 - b)}{N},$$

where K is the number of targets. For $K = 75$ this simplifies to $(74 + b)/149$, and for $K = 74$ to $(75 - b)/149$. Because the class imbalance is only one trial, the expected accuracy remains extremely close to 0.5 for any b (maximum deviation ≈ 0.003).

The variance of accuracy under this null is driven by trial-to-trial guessing variability. Since each trial is correct with probability approximately 0.5, the standard deviation of a single subject's accuracy is

$$\sqrt{\frac{0.5(1 - 0.5)}{149}} \approx 0.041.$$

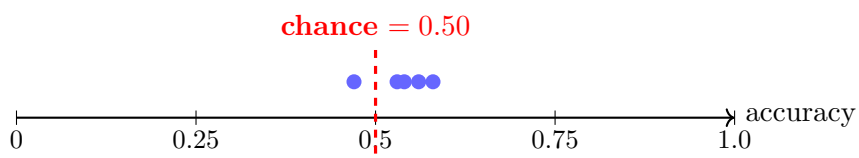
Thus, a single subject scoring 0.54 falls well within the range expected by chance. However, for a group of $n = 16$ subjects, the standard error of the mean decreases by a factor of $\sqrt{n}$:

$$\text{SE} = \frac{0.041}{\sqrt{16}} \approx 0.010.$$

Consequently, a group mean of 0.526 is substantially less likely to arise under the null than an individual deviation of similar magnitude.

We are interested to make a judgement about whether or not the subject responses are better explained by this null "guessing hypothesis" or by some other hypothesis, namely that subjects are able to do this task above chance.

Put graphically, does the cloud of points below fall above the mean in a reliable way that would be unlikely if subjects were just guessing? The red dashed line at 0.50 is the null value; the blue points are individual subjects;



2.2 The p-value

Probability of a result at least as extreme as the observed one, under the null hypothesis. Smaller p = more surprising under the null. The conventional cutoff is $p < 0.05$.

The typical intuition and story you can tell yourself: Pretend the null is true (subjects are guessing). Now imagine re-running the whole experiment a million times in that imaginary world where the null is true. A two-tailed p-value is the proportion of those imaginary experiments where the group mean is more than x percent away from the expected mean (either more the x percent below the mean or x percent above the mean). A one-tailed p-value is the same, except you are asking **either** roportion of those imaginary experiments where the group mean is x percent **greater** than the expected mean **or** the proportion where the group mean is x percent **less** than the expected mean, depending on the direction of your alternative hypothesis.

A p-value of $p = 0.05$ means *1 in 20 imaginary null-world experiments would look this extreme by accident.* A p-value of $p = 0.001$ means *1 in 1000 imaginary null-world experiments would look this extreme by accident.*

The one-tailed p-value is always half the two-tailed p-value, because it only counts one of those tails. It is tempting to use one-tailed tests because it automatically cuts your p-value in half, but you should only do it if you have a strong theoretical reason to expect the effect to go in one direction and not the other. In this experiment, we took an exploratory data analysis approach and we did not have a sttrong enough hypothesis to justify a one-tailed test. As a result, all of the analyses we ran were two-tailed. This is the safest way to make sure that we took all precautions to avoid false positives, and it is the most conservative approach.

A p-value is *not* the probability that the null is true, and it is *not* the probability that the effect is real. It is a statement about how often pure noise would mimic the observation.

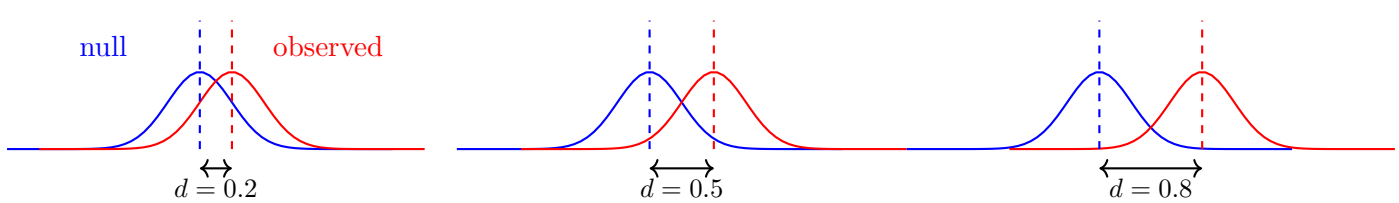
When we run many tests (e.g. 6 cells in Section A, 48 correlations in Section F) some will land below 0.05 by accident. The more tests you run, the more likely you are to see a false-postive. If you run enough tetsts, then you will expect to see one or more false-positive. If you roll a 20 sided die then you don't typically expect to roll a 4. But if you roll the die 1000 times then you expect to see *some* 4s. We correct for this using a conservative p-value correction known as Bonferroni correction. The Bonferroni correction divides our significance level $\alpha = 0.05$ by the number of tests; so if we run 6 tests then we take $\alpha = \frac{0.05}{6} = 0.0083$ as our new threshold for significance; if we run 48 tests then we take $\alpha = \frac{0.05}{48} \approx 0.001$ as our new threshold for significance.

2.3 Cohen's d

Effect size in units of pooled standard deviation. Rough rule of thumb: $|d| \approx 0.2$ small, 0.5 medium, 0.8 large.

The intuition: p-values conflate two things — how big the effect is, and how many subjects you ran. Run enough subjects and a tiny effect will clear $p < 0.05$; run too few and a large effect will miss it. Cohen's d strips out sample size and tells you the size of the gap between the observed mean and the null in units of how spread out the subjects are.

A concrete way to read it: $d = 1.0$ means the observed mean is one full standard deviation away from the null value. If subjects' scores formed a bell curve centered on chance, an observed group mean at $d = 1.0$ would sit out near the edge of that bell, in the region where only $\sim 16\%$ of single subjects naturally fall. The picture below shows three null/observed pairs at $d = 0.2, 0.5, 0.8$ — notice how much the two distributions still overlap even at $d = 0.8$.



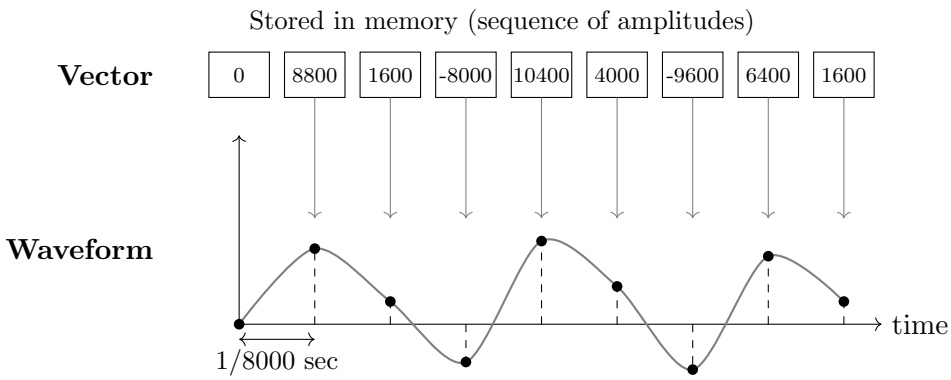
2.4 Cosine Similarity Classification Images

Our audio files are in PCM format, which stands for Pulse Code Modulation. This is a common way to digitally represent audio signals. They are 8 kHz, 16bit, mono, meaning that they have a sampling rate of 8000 samples per second, each sample is represented by 16 bits (2 bytes), and there

is only one audio channel (mono). An audio file in PCM format is simply a sequence of numbers stored in memory:

$$[x_1, x_2, x_3, \ldots, x_N].$$

Each number is the *amplitude* of the sound at a specific moment in time. For an 8 kHz file, the computer reads 8000 of these numbers per second and sends them to the speaker. If we plot these values against time and connect them, we obtain the waveform we hear.

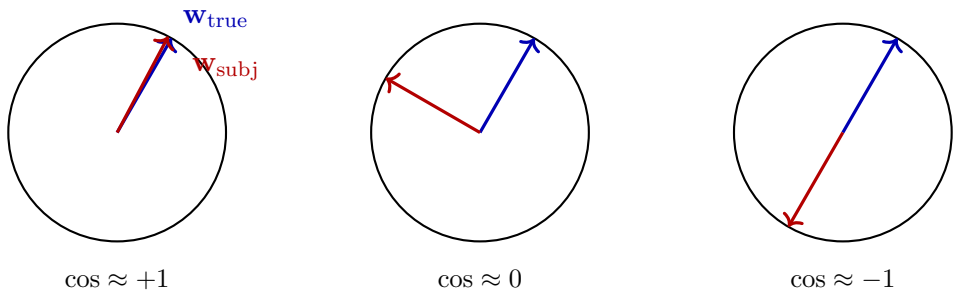


Intuition: the computer stores sound as a list of numbers. Each number says "how far the speaker should move" at that instant. Playing audio just means reading these numbers very quickly (8000 per second here) and turning them into motion. Plotting those numbers over time reveals the familiar waveform.

A classification image is also a long list of numbers, i.e. a vector. The cosine similarity between two vectors is the cosine of the angle between them in high-dimensional space. It ranges from $+1$ (identical direction) to 0 (orthogonal) to -1 (opposite direction). We use it to compare a subject's classification image to the true target waveform. A waveform or per-subject classification image is a long list of numbers, i.e. a vector. The cosine of the angle between two vectors is $+1$ if identical direction, 0 if orthogonal, -1 if opposite. We use it to compare a subject's classification image to the true target waveform.

The intuition: a subject's *classification image* is built by averaging the audio waveforms of all the trials they called "target" and subtracting the average of trials they called "distractor". If they were guessing, this difference is just averaged noise and points in a random direction. If they were actually picking up on the target, this difference should look something like the real target waveform — so the angle between their classification image and the true target should be small, and the cosine close to $+1$.

Why cosine and not raw dot product: cosine ignores how loud or contrasty the subject's classification image happens to be (its length) and only asks whether it points in the right direction. Two subjects with very different overall response biases can both have $\cos = +0.5$ if their classification images both *shape-match* the target.



2.5 d' (signal-detection sensitivity)

d' separates "can the subject tell target from distractor" from "does the subject lean toward saying yes or no". Two subjects can both score 0.55 accuracy, but one might be balanced (hits and correct rejections both around 0.55) while the other is biased (says "target" on almost every trial, racking up hits but also false alarms). d' rewards the first and not the second.

Concretely, $d' = z(\text{hit rate}) - z(\text{false-alarm rate})$. Null value is 0. As a sanity check, an accuracy of 0.526 with no bias corresponds to $d' \approx 0.13$, matching the Oberlin full-sentence row in Section E.

2.6 $\beta_{\textbf{truetarget}}$ (regression slope on the true target)

A logistic regression of the subject's per-trial response on the trial-by-trial dot product between the stimulus waveform and the true target waveform. A positive slope means *the more the trial's audio resembled the true target, the more likely the subject was to call it "target"*. Null value is 0. This is a direct test of target-driven responding, with units of log-odds per unit of target-similarity.

3 Results

3.1 Section A: Accuracy vs chance

Section A (`A_accuracy_vs_chance.csv`)

block	group	n	mean	95% CI	Wilcoxon p	d
full_sentence	UChicago	10	0.504	$[0.479, 0.530]$	0.869	+0.09
full_sentence	Oberlin	16	0.526	$[0.513, 0.538]$	0.0037	+0.99
full_sentence	Unified	26	0.518	$[0.504, 0.530]$	0.019	+0.50
imagined_sentence	UChicago	10	0.490	$[0.464, 0.515]$	0.611	−0.23
imagined_sentence	Oberlin	16	0.510	$[0.495, 0.526]$	0.277	+0.31
imagined_sentence	Unified	26	0.502	$[0.488, 0.517]$	0.517	+0.06

Bonferroni-corrected threshold across the six tests is $\alpha = 0.05/6 \approx 0.0083$. Oberlin full-sentence ($p = 0.0037$) clears it; the Unified full-sentence cell ($p = 0.019$) does not.

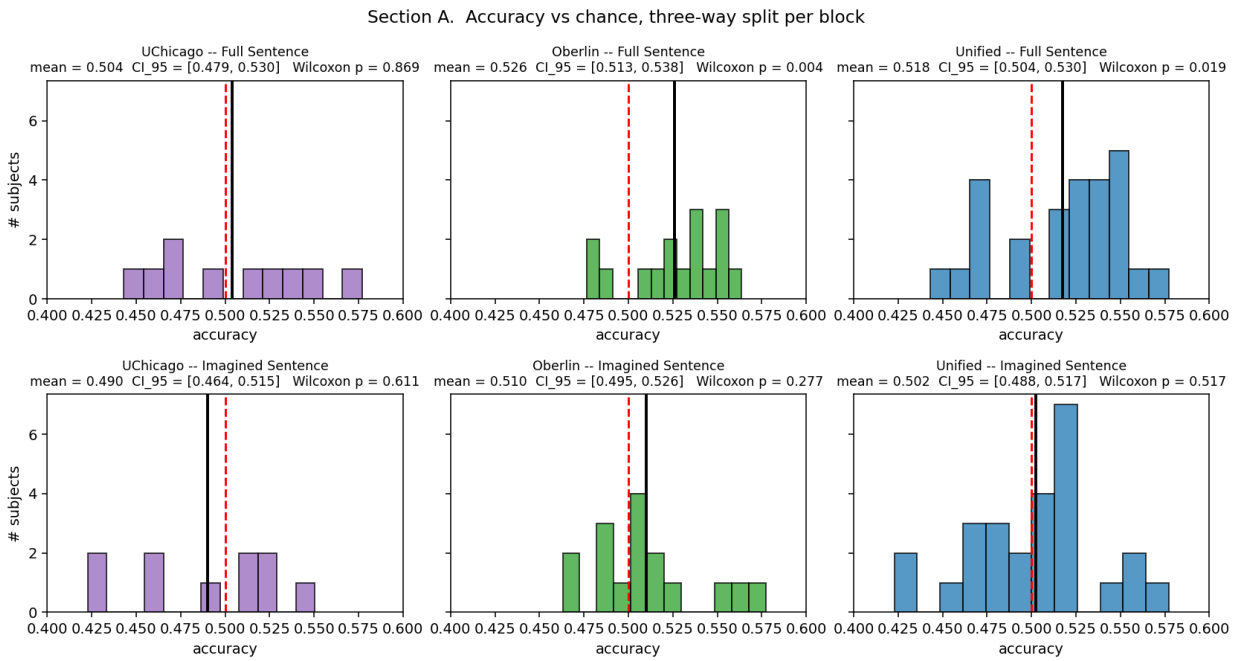


Figure 1: Per-subject accuracy by (block, group). Red dashed = chance (0.50); black solid = group mean. Title shows the bootstrap CI on the mean and the Wilcoxon p vs. chance.

3.2 Section B: Full vs Imagined (paired)

Section B (`B_full_vs_imagined.csv`)

group	n	mean(F)	mean(I)	mean diff	paired Wilcoxon p	perm p	d
UChicago	10	0.504	0.490	+0.0141	0.543	0.508	+0.22
Oberlin	16	0.526	0.510	+0.0159	0.209	0.198	+0.35
Unified	26	0.518	0.502	+0.0152	0.157	0.158	+0.29

The mean difference is +0.0152 pooled and positive in both sites. No paired test reaches $p < 0.05$.

3.3 Section C: Site comparison

Section C (`C_site_comparison.csv`)

block	n_O	n_U	mean$_O$	mean$_U$	diff	MWU p	Welch p	d
full_sentence	16	10	0.526	0.504	+0.0220	0.145	0.173	+0.65
imagined_sentence	16	10	0.510	0.490	+0.0201	0.475	0.231	+0.54

Oberlin is numerically higher than UChicago in both blocks by ≈ 0.02. Neither between-site comparison reaches $p < 0.05$.

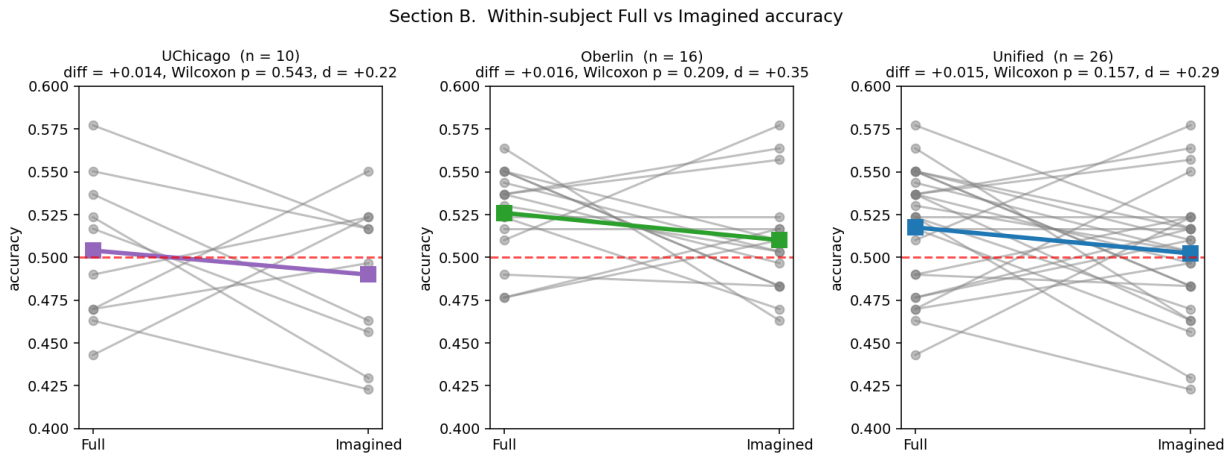


Figure 2: Per-subject lines connecting Full to Imagined accuracy in each group; coloured heavy line is the group mean.

3.4 Section D: Block order

Section D selected rows (`D_block_order.csv`)

block	group	n_{B1}	n_{B2}	mean(B1)	mean(B2)	MWU p	d
full_sentence	Oberlin	5	11	0.519	0.529	0.493	+0.35
imagined_sentence	Oberlin	11	5	0.500	0.532	0.078	+1.03

In the Oberlin imagined block, subjects who completed it second ($n_{B2} = 5$) averaged 0.532, vs 0.500 for those who completed it first ($n_{B1} = 11$); MWU $p = 0.078$, $d = +1.03$. All other block-order cells have $p > 0.4$.

3.5 Section E: Convergent metrics, full-sentence block

Section E (`E_convergent_full_sentence.csv`)

group	metric	mean	95% CI	Wilcoxon p	d
Oberlin	d′	+0.137	[+0.069, +0.200]	0.0013	+0.99
Oberlin	accuracy	+0.526	[+0.513, +0.538]	0.0037	+0.99
Oberlin	cos(CI, true)	+0.056	[+0.029, +0.083]	0.0013	+1.00
Oberlin	beta_truetarget	+0.232	[+0.118, +0.340]	0.0027	+0.99
UChicago	d′	+0.021	[−0.106, +0.153]	0.846	+0.09
UChicago	cos(CI, true)	−0.002	[−0.059, +0.056]	1.000	−0.02
UChicago	beta_truetarget	+0.042	[−0.159, +0.248]	0.846	+0.12
Unified	d′	+0.093	[+0.023, +0.159]	0.013	+0.51
Unified	cos(CI, true)	+0.034	[+0.004, +0.063]	0.025	+0.43
Unified	beta_truetarget	+0.159	[+0.047, +0.267]	0.016	+0.55

At Oberlin all four metrics yield $p \leq 0.0037$ and $d \approx 1.0$. At UChicago all four are non-significant with $|d| \leq 0.12$. Pooled, all four yield $p \in [0.013, 0.025]$ and $d \in [0.43, 0.55]$.

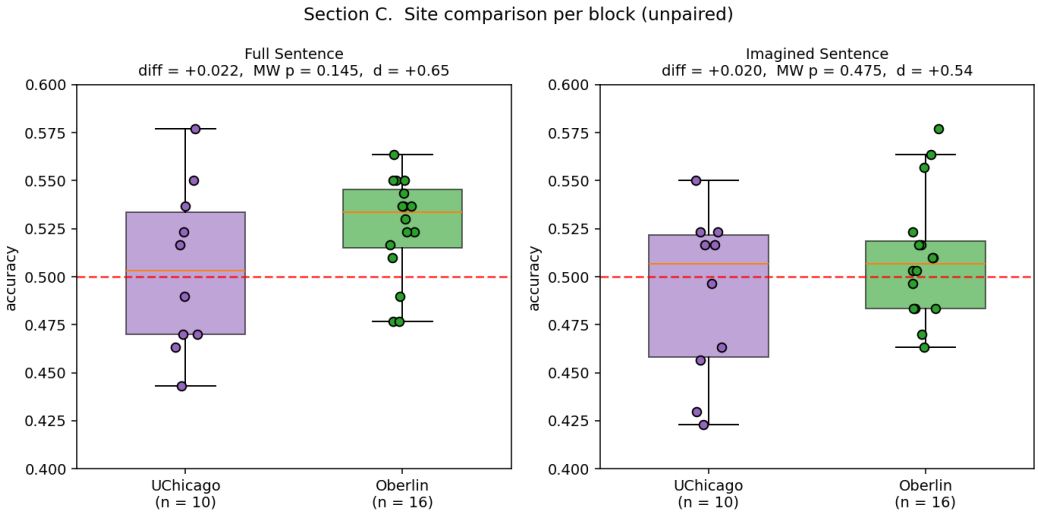


Figure 3: Per-block boxplots, Oberlin vs. UChicago, with overlaid jittered subject points.

3.6 Section F: Questionnaire correlations

48 Spearman correlations (8 questionnaires $\times$ 2 blocks $\times$ 3 groups). Bonferroni threshold $p < 0.001$. None survive correction. Tables: `F_questionnaire_spearman.csv`; figures: `F_quest_vs_accuracy_full_sentence.` `F_quest_vs_accuracy_imagined_sentence.png`.

4 Further analyses available with the existing data

4.1 Trial-by-trial mixed-effects modelling

The current per-subject summaries collapse ∼150 trials per block into a single mean. A mixed-effects logistic regression on raw trials, with subject as a random effect, would use the within-subject trial variance directly.

4.2 Bayesian re-analysis of trending comparisons

For comparisons with $0.05 < p < 0.20$ (Section B Full–Imagined paired, Section C site differences), Bayes factors quantify the evidence ratio between H_1 and H_0 rather than only rejecting H_0.

4.3 Within-subject correlations between convergent metrics

The four Section E metrics are tested independently. Computing the per-subject correlation matrix between d', accuracy, $\cos(\mathrm{CI}, \mathrm{true})$, and $\beta_{\mathrm{truetarget}}$ tests whether they are driven by a single latent perceptual ability.

4.4 Spectral classification image

A 32-band spectral analogue of the time-domain classification image: for each subject, fit per-band weights and compute the cosine with the true target's per-band spectrum.

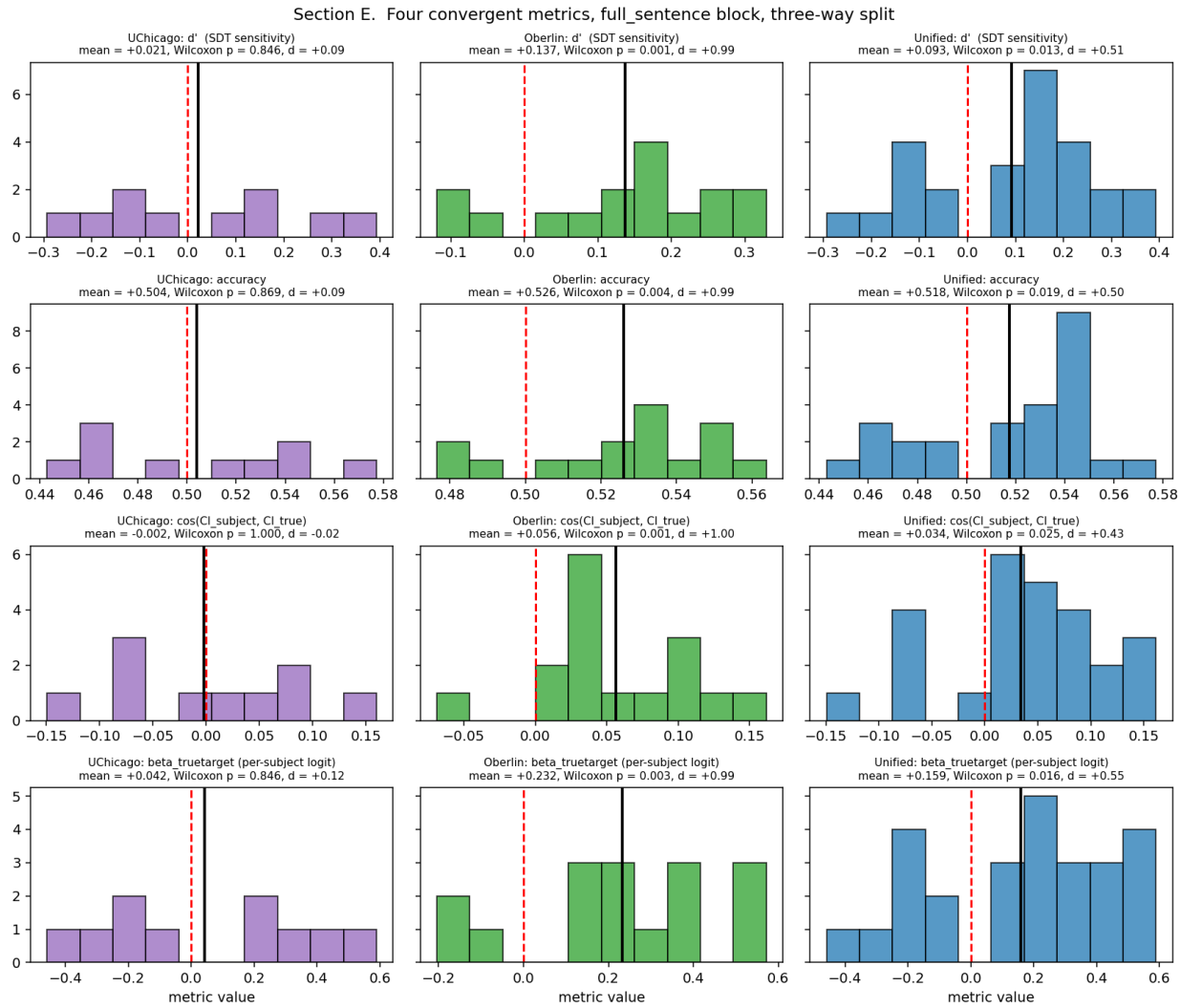
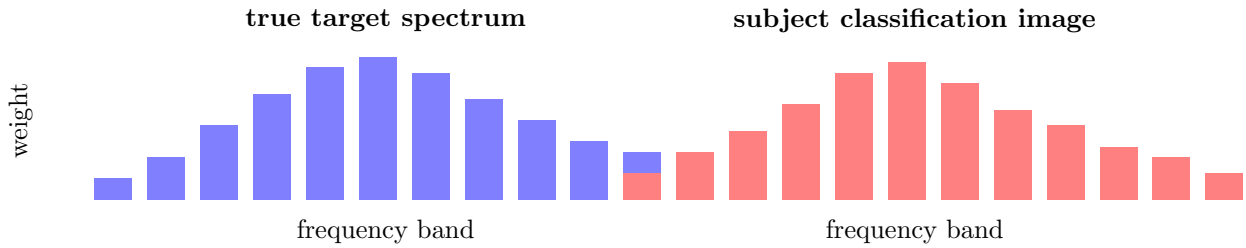


Figure 4: Each row is a metric, each column a group. Red dashed = null value (0 or 0.5); black solid = observed mean.



4.5 Block-order follow-up in the Oberlin imagined block

Trial-resolved analysis of the $n_{B2} = 5$ vs $n_{B1} = 11$ split: e.g. within-block accuracy time course, and whether the +0.032 mean difference is concentrated in early or late trials of the block.

4.6 Stimulus-level analyses

Per-stimulus accuracy across the audio set, to test whether particular target/distractor pairs drive the Oberlin full-sentence effect. Building blocks for this exist in `stimuli-analyses/` (classification-image, spectral logistic, etc.).